GeoFM evaluation framework

One configuration connects data preparation, model training and evaluation.

YAML Dataset & splits / Modalities / Encoder & head / Training / Evaluation

01 DATA

Images + segmentation labels

CSV splits: train / validation / test

S1, S2 or paired S1 + S2

Training: aligned random crops

Normalization: training statistics
or fixed pretraining statistics

02 MODEL

CROMA or TerraMind

Local pretrained encoder

Tokens to spatial feature maps

Linear or FCN segmentation head

Upsampled pixel predictions
Current probes: encoder frozen

03 TRAIN & EVALUATE

Train on the training split

Validate every N epochs

Select best validation checkpoint

Reload selected weights, then test

Validation/test: tiled inference
Test and Bolivia reported separately*

Independent random streams

Shared model interface

Selection uses validation only

ONE FOLDER PER RUN

Configuration, seeds & hashes
Best / last checkpoints

METRICS

Loss, mIoU, per-class IoU, precision & recall
AW / PW / FW: per-chip and pooled IoU*

REPORTS

JSON summaries & history
Training and test plots

* Sen1Floods11-specific outputs, when enabled. Ignored labels are excluded from metric calculations.